# Assignment 5

Q1.Run an Apache Pod using image httpd. Verify that the Pod is running

```
mojorojo@mojorojo-vmware:~/k8s$ kubectl run my-apache --image httpd
pod/my-apache created
mojorojo@mojorojo-vmware:~/k8s$ kubectl get pods
NAME        READY   STATUS              RESTARTS   AGE
my-apache   0/1     ContainerCreating   0          5s
mojorojo@mojorojo-vmware:~/k8s$ kubectl get pods w
Error from server (NotFound): pods "w" not found
mojorojo@mojorojo-vmware:~/k8s$ kubectl get pods b
Error from server (NotFound): pods "b" not found
mojorojo@mojorojo-vmware:~/k8s$ kubectl get pods -b
error: unknown shorthand flag: 'b' in -b
See 'kubectl get --help' for usage.
mojorojo@mojorojo-vmware:~/k8s$ kubectl get pods -w
NAME        READY   STATUS    RESTARTS   AGE
my-apache   1/1     Running   0          22s
^C
```

Q2. Create a Deployment using image httpd with 2 replicas. Verify the Pods created by the Deployment.

```
mojorojo@mojorojo-vmware:~/k8s$ kubectl run pod my-http --image httpd --replicas 2
error: unknown flag: --replicas
See 'kubectl run --help' for usage.
mojorojo@mojorojo-vmware:~/k8s$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   11m   v1.35.1
mojorojo@mojorojo-vmware:~/k8s$ kubectl create deployment my-httpd --image nginx
deployment.apps/my-httpd created
mojorojo@mojorojo-vmware:~/k8s$ kubectl scale deployment my-httpd --replicas 2
deployment.apps/my-httpd scaled
mojorojo@mojorojo-vmware:~/k8s$ kubectl get rs
NAME                  DESIRED   CURRENT   READY   AGE
my-httpd-7f7f649f68   2         2         2       29s
```

Q3. Increase the number of replicas to 4 and then decrease it to 1. Verify the changes

```
mojorojo@mojorojo-vmware:~/k8s$ kubectl create deployment my-httpd --image nginx
deployment.apps/my-httpd created
mojorojo@mojorojo-vmware:~/k8s$ kubectl scale deployment my-httpd --replicas 2
deployment.apps/my-httpd scaled
mojorojo@mojorojo-vmware:~/k8s$ kubectl get rs
NAME                   DESIRED   CURRENT   READY   AGE
my-httpd-7f7f649f68    2         2         2       29s
mojorojo@mojorojo-vmware:~/k8s$ kubectl scale deployment my-httpd --replicas 4
deployment.apps/my-httpd scaled
mojorojo@mojorojo-vmware:~/k8s$ kubectl get rs
NAME                   DESIRED   CURRENT   READY   AGE
my-httpd-7f7f649f68    4         4         4       2m8s
mojorojo@mojorojo-vmware:~/k8s$ kubectl scale my-httpd --repljicas 1
error: unknown flag: --repljicas
See 'kubectl scale --help' for usage.
mojorojo@mojorojo-vmware:~/k8s$ kubectl scale my-httpd --replicas 1
error: the server doesn't have a resource type "my-httpd"
mojorojo@mojorojo-vmware:~/k8s$ kubectl scale deployment my-httpd --replicas 1
deployment.apps/my-httpd scaled
mojorojo@mojorojo-vmware:~/k8s$ kubectl get rs
NAME                   DESIRED   CURRENT   READY   AGE
my-httpd-7f7f649f68    1         1         1       3m10s
mojorojo@mojorojo-vmware:~/k8s$
```

---

Q4.

Create a YAML file to define a Deployment using image

Apply the YAML file and verify the Deployment.

httpd

with 3 replicas

```
mojorojo@mojorojo-vmware:~/yamldeployment$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-deploy
spec:
  replicas: 3
  selector:
    matchLabels:
      app: my-dep1
  template:
    metadata:
      labels:
        app: my-dep1
    spec:
      containers:
      - name: deploy-demo
        image: httpd
        ports:
        - containerPort: 80
```

```
mojorojo@mojorojo-vmware:~/yamldeployment$ kubectl get deployment
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
my-deploy     3/3     3            3           24s
my-httpd      1/1     1            1           2d
```

---

Q5. Expose the Deployment using:

ClusterIP Service

```
mojorojo@mojorojo-vmware:~/yamldeployment$ kubectl expose deployment my-deploy  --name=httpd-clusterip --port=80 --target-po
=80 --type=ClusterIP
service/httpd-clusterip exposed
mojorojo@mojorojo-vmware:~/yamldeployment$ kubectl get svc
NAME                 TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
httpd-clusterip      ClusterIP      10.107.31.59     <none>        80/TCP         6s
httpd-loadbalancer   LoadBalancer   10.111.110.214   <pending>     80:32247/TCP   3m13s
httpd-nodeport       NodePort       10.102.46.92     <none>        80:32297/TCP   6m25s
kubernetes           ClusterIP      10.96.0.1        <none>        443/TCP        2d1h
```

NodePort Service

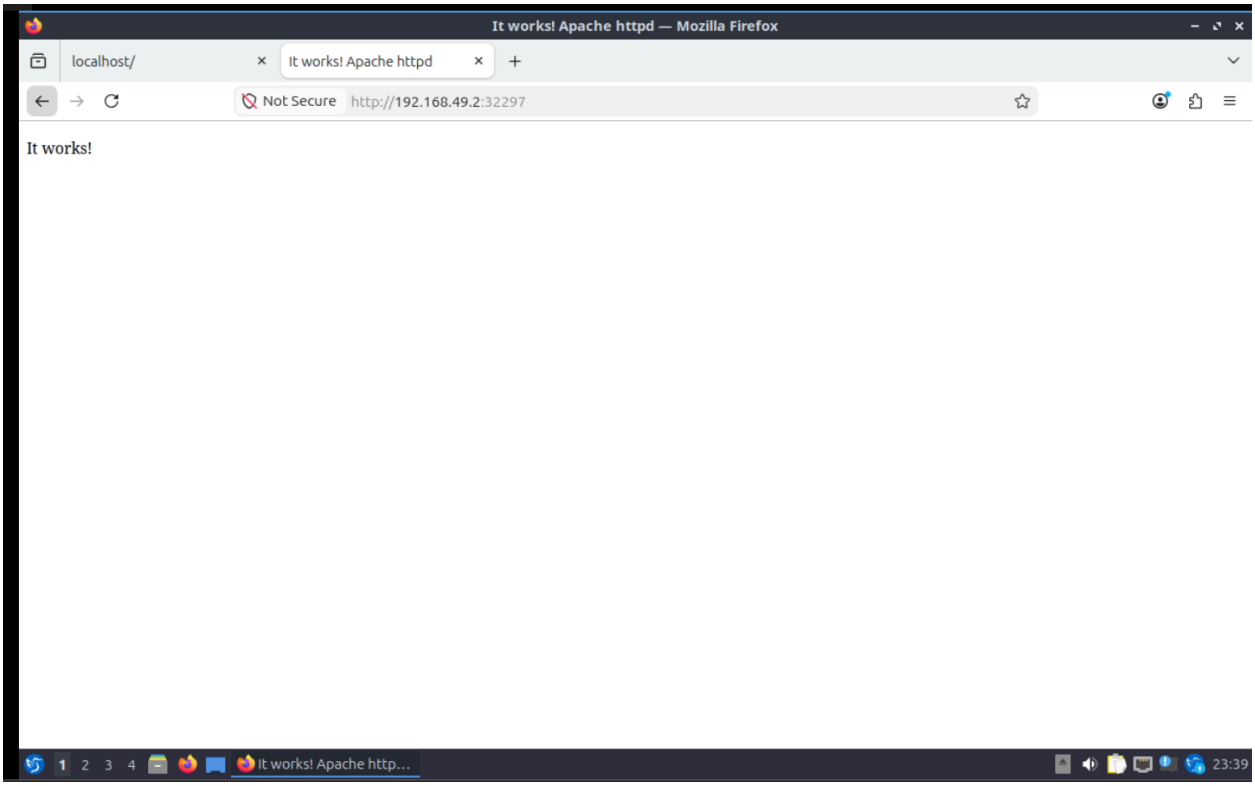
```
mojorojo@mojorojo-vmware:~/yamldeployment$ kubectl expose deployment my-deploy  --name=httpd-nodeport --port=80 --target-port=
80 --type=NodePort-node
The Service "httpd-nodeport" is invalid: spec.type: Unsupported value: "NodePort-node": supported values: "ClusterIP", "Extern
alName", "LoadBalancer", "NodePort"
mojorojo@mojorojo-vmware:~/yamldeployment$ kubectl expose deployment my-deploy  --name=httpd-nodeport --port=80 --target-port=
80 --type=NodePort
```

## LoadBalancer Service

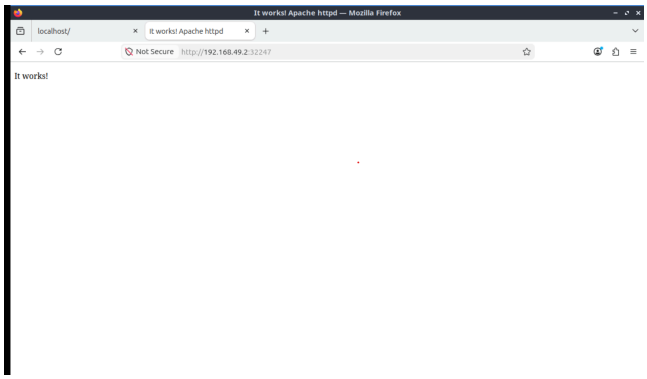
```
mojorojo@mojorojo-vmware:~/yamldeployment$ kubectl get svc
NAME             TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
httpd-nodeport   NodePort    10.102.46.92   <none>        80:32297/TCP   118s
kubernetes       ClusterIP   10.96.0.1      <none>        443/TCP        2d1h
mojorojo@mojorojo-vmware:~/yamldeployment$ kubectl expose deployment my-deploy  --name=httpd-loadbalancer --port=80 --target-p
ort=80 --type=loadbalancer
The Service "httpd-loadbalancer" is invalid: spec.type: Unsupported value: "loadbalancer": supported values: "ClusterIP", "Ext
ernalName", "LoadBalancer", "NodePort"
mojorojo@mojorojo-vmware:~/yamldeployment$ kubectl expose deployment my-deploy  --name=httpd-loadbalancer --port=80 --target-p
ort=80 --type=LoadBalancer
service/httpd-loadbalancer exposed
```

## Verify all services

## NODE-port verify



## Load-Balancer




## ALL

```
mojorojo@mojorojo-vmware:~/yamldeployment$ kubectl get svc
NAME                  TYPE           CLUSTER-IP       EXTERNAL-IP    PORT(S)        AGE
httpd-clusterip       ClusterIP      10.107.31.59     <none>         80/TCP         67s
httpd-loadbalancer    LoadBalancer   10.111.110.214   <pending>      80:32247/TCP   4m14s
httpd-nodeport        NodePort       10.102.46.92     <none>         80:32297/TCP   7m26s
kubernetes            ClusterIP      10.96.0.1        <none>         443/TCP        2d1h
```